

Land Acquisition Early Warning System (EWS)

Machine Learning Architecture, Model Evaluation & Live Dataset Demonstration

1. Executive Summary & Objective

Large-scale national infrastructure projects (highways, railways, energy corridors, industrial nodes) frequently experience severe multi-crore cost escalation and completion delays primarily driven by bottlenecks in **Land Acquisition** and **Rehabilitation & Resettlement (R&R)**. The Land Acquisition Early Warning System (EWS) is an advanced, explainable predictive AI model built to monitor longitudinal project snapshots and forecast critical completion delays.

Primary Prediction Target: `delayed_gt_90_days` (Binary classification: whether completion delay exceeds 90 days).

Operational Objective: Detect delay precursors 3 to 12 months in advance, equipping administrative authorities with granular, explainable root causes for targeted fast-track resolution.

2. Dataset Architecture & Leakage Prevention

The model is trained and validated on a master database containing longitudinal monitoring snapshots across infrastructure initiatives:

Attribute	Specification / Dimension	Operational Details
Master Dataset File	land_acquisition_master.csv	Primary project monitoring master repository
Total Snapshot Records	30,000 records	Regular milestone / quarterly monitoring events
Unique Projects	6,000 distinct projects	5 longitudinal monitoring snapshots per project
Input Feature Space	46 clean predictive features	42 continuous/numerical + 4 categorical features
Leakage Quarantined	4 target-proxy columns	Excluded: actual delay days, delay severity, most likely delay stage

3. Zero-Overlap Group Partitioning Strategy

Standard random cross-validation causes optimistic data leakage when evaluating multi-snapshot project data. To guarantee real-world generalization to new, unseen infrastructure projects, the system utilizes **GroupShuffleSplit** partitioned strictly on `project_id`:

Training Cohort (60%): 3,600 unique projects (18,000 snapshots) used exclusively for model training.

Validation Cohort (20%): 1,200 unique projects (6,000 snapshots) used for hyperparameter tuning & threshold optimization.

Test Cohort (20%): 1,200 unique projects (6,000 snapshots) reserved strictly for final, unbiased performance verification.

Overlap Verification: $\text{Train} \cap \text{Validation} = 0$, $\text{Train} \cap \text{Test} = 0$, $\text{Validation} \cap \text{Test} = 0$ (100% independent project groups).

4. Feature Taxonomy (The 11 Operational Dimensions)

The model analyzes 46 features capturing the complete multi-disciplinary lifecycle of land acquisition:

Domain	Count	Key Included Indicators & Metrics
Context & Geography	3	state, district, project_type (Highways, Railways, Energy, Ports)
Project Scale	3	project_cost, project_length_km, land_required_hectares
Land Composition	5	land_acquired_pct, land_pending_pct, private_land_pct, government_land_pct, forest_land_pct
Affected Population	3	affected_families, affected_landowners, vulnerable_households
Compensation & Finance	6	compensation_awarded, compensation_paid, compensation_pending, compensation_pending_pct, compensation_dispute_count, average_compensation_delay_days
Legal & Litigation	4	legal_case_count, court_case_count, arbitration_case_count, ownership_dispute_count
Administrative Delays	5	notification_delay_days, approval_delay_days, survey_delay_days, document_completion_pct, interdepartmental_pending_count
Rehabilitation & Resettlement	4	rr_required, rr_completion_pct, families_relocated_pct, rr_grievances
Possession & ROW	3	possession_pct, row_available_pct, encumbrance_free_pct
Stakeholder Grievances	3	public_objection_count, unresolved_grievances, stakeholder_response_rate
Historical & Temporal	7	district_avg_resolution_days, agency_avg_delay_days, previous_project_delay_rate, days_since_notification, days_since_last_update, days_in_current_stage, acquisition_stage

5. Preprocessing Pipeline & ML Architecture

Data Preprocessing Pipeline: Structured using scikit-learn's ColumnTransformer:

- *Numerical Features (42):* Imputed with median values followed by StandardScaler().
- *Categorical Features (4):* Imputed with mode followed by OneHotEncoder(handle_unknown='ignore').

Champion Model — Extreme Gradient Boosting (XGBoost):

A regularized histogram-based decision tree ensemble designed to capture non-linear interactions across legal and financial hold-ups:

Hyperparameter	Configured Value	Engineering Rationale
Objective	binary:logistic	Outputs calibrated probabilities P(Delay > 90 days)
Tree Method	hist	Fast histogram-based binning for optimal memory efficiency
Number of Estimators	500 trees	Deep ensemble capacity with conservative learning rate
Learning Rate (eta)	0.03	Gradual gradient convergence to eliminate overfitting
Max Depth	6	Captures multi-way interactions (e.g., litigation + pending R&R;)
Subsample & ColSample	0.8 / 0.8	Stochastic row & column subsampling for generalization variance
L1 Regularization (alpha)	0.1	Sparse feature selection to suppress uninformative noise
L2 Regularization (lambda)	1.0	L2 ridge penalty on leaf weights to avoid extreme predictions

6. Operational Decision Framework & Risk Tiers

The model's raw probability output is translated directly into operational decision tiers for nodal project officers:

Risk Tier	Probability Range	Recommended Administrative Action
LOW RISK	< 40%	Normal operational track. Milestone monitoring on standard schedule.
MODERATE RISK	40% – 59%	Early warnings detected. Monthly inter-departmental review recommended.
HIGH RISK	60% – 79%	Significant compensation/dispute backlog. Fast-track administrative intervention.
CRITICAL RISK	≥ 80%	Severe delay imminent (> 90 days). Escalate to Ministry / Special Task Force.

7. Explainable AI (XAI) with SHAP

Government stakeholders require full auditability and transparency. The EWS integrates **SHAP (SHapley Additive exPlanations)** via shap.TreeExplainer:

- Global Policy Insights:** Identifies nationwide systemic bottlenecks across thousands of projects (e.g., compensation dispute rate, forest land clearance lags, and agency average delay rate).
- Project-Specific Root Cause Attribution:** For any specific project, calculates local Shapley values that rank the exact factors inflating delay probability, enabling officers to tackle the exact root cause rather than guessing.

8. Benchmark Performance & Evaluation Metrics (Validation Cohort)

Evaluated on 1,200 independent projects (6,000 snapshots) with optimal threshold \$T=0.50\$:

Evaluation Metric	Logistic Regression Baseline	XGBoost Champion Model
Accuracy	90.80%	91.37%
Precision	95.75%	93.42%
Recall	90.49%	93.92%
F1-Score	93.04%	93.67%
ROC-AUC	0.9745	0.9741
PR-AUC (Avg Precision)	0.9883	0.9880

9. Live Demonstration & Model Prediction Outputs

The following real-time outputs were generated directly by executing the model on `land_acquisition_master.csv`. These demonstrate both batch classification performance and project-level explainable root-cause diagnosis.

9.1 Real Model Confusion Matrix (XGBoost Validation Set — 6,000 Samples)

Actual Class \ Predicted	Predicted: On-Time / Minor (0)	Predicted: Delayed > 90 Days (1)
Actual: On-Time / Minor (0)	1,651 (True Negatives)	270 (False Positives)
Actual: Delayed > 90 Days (1)	248 (False Negatives)	3,831 (True Positives)

9.2 Batch Inference Demonstration on Real Project Snapshots

Demonstration of model inference across diverse Indian states and infrastructure categories from `land_acquisition_master.csv`:

Project ID	State	Sector	Cost (Cr)	Delay Prob	Assigned Risk Tier	Actual Status
P0009	Gujarat	Industrial Corridor	Rs. 266.95	35.06%	LOW	On-Time (0)
P0015	Karnataka	Highway	Rs. 166.82	0.18%	LOW	On-Time (0)
P0016	Madhya Pradesh	Highway	Rs. 126.43	91.35%	CRITICAL	Delayed > 90d (1)
P0018	Maharashtra	Highway	Rs. 196.53	0.63%	LOW	On-Time (0)
P0024	Andhra Pradesh	Highway	Rs. 278.05	99.90%	CRITICAL	Delayed > 90d (1)
P0030	Telangana	Railway	Rs. 81.06	99.40%	CRITICAL	Delayed > 90d (1)
P0032	Haryana	Highway	Rs. 151.35	25.83%	LOW	On-Time (0)

9.3 Explainable AI Case Studies (Root Cause Attribution)

Case Study	Key Project Metrics	Top SHAP Impact Drivers & Decision Rationale
Case A: Project P0009 (Gujarat Industrial Corridor) Risk: LOW (35.06%)	- Cost: Rs. 266.95 Cr - Land: 19.72 Ha - Stage: Compensation Disbursement - Pending Comp: 77.01% - Legal Cases: 0	Strong Protective Signals: - stakeholder_response_rate (SHAP: -1.50): Proactive community dialogue. - days_since_notification (SHAP: -1.12): Smooth initial timeline. - ownership_dispute_count (SHAP: -1.02): Clear land titles with zero disputes. Primary Alert Driver: Pending compensation disbursement (+1.08).
Case B: Project P0016 (Madhya Pradesh Highway) Risk: CRITICAL (91.35%)	- Cost: Rs. 126.43 Cr - Land: 96.98 Ha - Stage: Joint Measurement - Pending Comp: 88.89% - Legal Cases: 4 (Court: 2) - Public Objections: 9	Severe Delay Escalation Drivers: - legal_case_count & court_case_count (High Positive Impact): 4 active court litigations stalling survey. - public_objection_count (High Impact): 9 formal stakeholder objections. - compensation_pending_pct: 88.89% compensation unawarded. Prescribed Action: Immediate fast-track legal tribunal settlement.

Conclusion for Faculty Evaluation: The EWS model proves not only statistical robustness (> 91% accuracy, > 0.97 ROC-AUC), but also operational readiness. By pairing calibrated risk probabilities with local SHAP feature attributions, project directors can detect high-risk infrastructure bottlenecks months before physical construction is disrupted.